Tangkhul Village Directory

Google AI Studio Prompt — Interactive, Editable PWA

Copy this prompt into Google AI Studio and provide your existing village directory document as source data.

1. Main Objective

Build or upgrade a fully functional, installable Progressive Web App (PWA) using my existing Tangkhul Village Directory document as the source of data. Users must be able to view, add, edit, update, search, filter, and delete village records. All buttons, forms, tables, filters, menus, and dashboard statistics must work. Do not create a static design or demo-only interface.

2. Design and User Interface

Use a professional, clean light-mode design: white background, subtle blue accents, dark readable text, simple tables and forms, clean spacing, and mobile-first responsiveness for Android phones, tablets, laptops, and desktops. Use minimal borders and subtle shadows. Prefer text labels over unnecessary icons. Avoid decorative icons, excessive animations, gradients, and complicated layouts. Make the app easy for people with limited technical knowledge.

3. Navigation Menu

Create a collapsible sidebar on desktop and a slide-out menu on mobile. Menu items: Dashboard; Village Directory; Add New Village; Data Import / Export; Settings. Use simple text labels, highlight the selected page, and allow the menu to expand and collapse. On mobile, use a simple button labelled “Menu”. Avoid unnecessary icons.

4. Dashboard

Show live summary cards for total villages; villages in North, East, West, and South; total recorded votes; villages with missing information; and recently updated records. Include simple charts for villages by region and recorded votes by region. Recalculate statistics whenever records are added, edited, or deleted. Do not count blank or unverified vote values as zero; distinguish missing values from actual zero values.

5. Village Directory

Display records in a searchable table with columns: Sl. No.; Name of Village; Region; Longphang; No. of Votes; President; Contact No.; Misc.; Actions. Search by village name, region, Longphang, president, and contact number. Filter by North, East, West, South, missing information, and recently updated. Sort by village name, region, and vote count. Provide table and mobile-friendly list views, plus pagination or efficient scrolling for large datasets.

6. Add and Edit Records

Create a form with Village Name (required), Region (North/East/West/South), Longphang (selectable or editable), No. of Votes (optional number), President (optional), Contact Number (optional), and Miscellaneous Notes (optional). Support adding, editing, saving, cancelling, and deleting after confirmation. Validate required fields and show clear success/error messages. Update serial numbers and dashboard statistics automatically. Warn about possible duplicate village names but never merge them automatically. Allow incomplete records to be saved. When editing, prefill existing values and prevent accidental blank overwrites.

7. Data Storage and Persistence

Use IndexedDB for local structured data storage. Data must remain available after refresh, closing, and reopening the app. Load saved records at startup, show when changes are saved, handle storage errors, use stable record IDs rather than serial numbers, and store a last-updated timestamp. Do not silently overwrite records. Include JSON backup and restore. Explain that clearing browser site data can erase local records and that browser storage is not encrypted by default.

8. Import Existing Data

Import the actual records from the provided Tangkhul Village Directory document, preserving village names and spelling, region, Longphang headings, recorded vote counts, president names, contact numbers, and miscellaneous notes only where present. Preserve the regional groupings: North — Ato; East — Zingsho; West — Zingtun; South — Aze. Keep unknown fields blank; preserve vote counts exactly; never treat missing votes as zero; do not invent values; exclude removed financial amounts/payment details; flag ambiguous or potential duplicate records for manual review. If direct document parsing is not reliable, offer CSV/JSON import and clearly report records that could not be imported confidently.

9. Import and Export

Provide a Data Import / Export page. Support CSV export (Excel-compatible), JSON export, CSV import, JSON backup restore, backup before bulk changes, preview before confirming, duplicate detection, validation errors, and a choice to skip duplicates or update matching records. Show a summary of added, updated, skipped, and failed records. Never silently erase or overwrite the existing database during import.

10. PWA Requirements

Include a valid web app manifest, service worker, appropriate app icons, standalone display mode, and installation support for compatible Android and desktop browsers. After initial setup, support offline app loading, offline viewing and editing, and reliable local persistence. Show offline/online status and a visible backup reminder if no recent backup exists. Core record management must not depend on an online database or external service. Explain HTTPS deployment and Android installation.

11. Privacy and Data Protection

Treat village contact information as private operational data. Do not send data to external servers without explicit configuration and consent. Do not add analytics or tracking. Do not expose records publicly by default. Do not claim local storage is encrypted. Provide backup and restore. Keep any future cloud synchronization separate and require authentication, access control, and conflict handling.

12. Settings

Include only simple settings: default region filter, table/list view, backup and restore, data validation preferences, app information, and clear-all-data protected by a strong confirmation and backup warning. Avoid unnecessary settings.

13. Technical Requirements

Use a suitable modern frontend such as React with TypeScript and clean reusable components. Use IndexedDB for persistent local storage. Implement validation, state management, error handling, and accessible controls. Use a lightweight charting solution or simple native charts. Keep the app fast on mid-range Android phones. Ensure no nonfunctional buttons, placeholder interactions, hardcoded dashboard totals, fake records, data loss when navigating, or accidental deletion/overwriting. Include keyboard navigation and readable contrast, plus loading, empty, error, and saved states.

14. Required Deliverables

Generate the complete working project, not just a design preview. Include all source code and configuration files, PWA manifest, service worker, IndexedDB implementation, initial dataset from the supplied document, working add/edit/delete/search/filter/sort, import/export/backup/restore, installation and deployment instructions, and a README explaining how to run, build, deploy, and maintain the application.

15. Final Testing Instruction

Prioritize simplicity, reliability, data accuracy, and ease of use. Build around the real village directory data. A user should be able to open a village, change information, press Save, and see the update everywhere. Before finishing, test editing an existing record, adding and deleting a village, refreshing, offline use, exporting, and restoring a backup. Do not stop after generating the interface; implement and verify the complete working app.